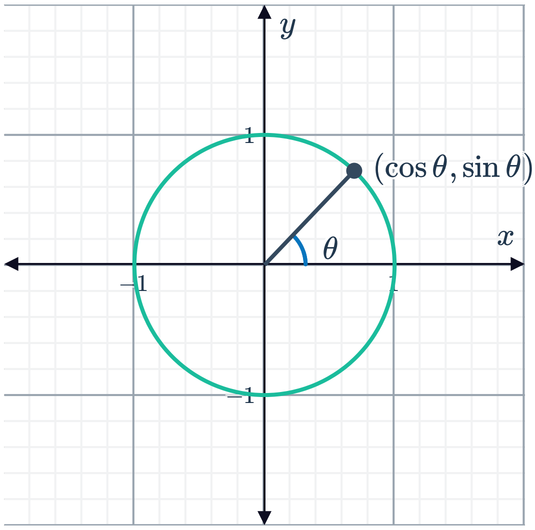
Graphs of trigonometric functions

1 The value of $\sin\theta$ can be represented on the given xy-plane.

a Consider the curve $y = \sin\theta$:
Complete the following:

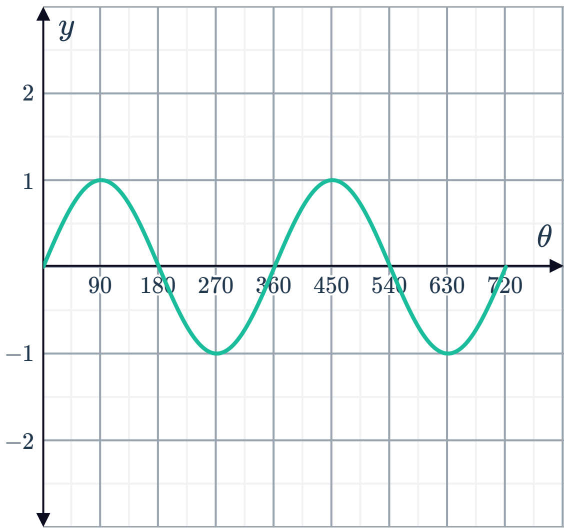


i A rotation of 380° has the same rotation as the acute angle with measure ▯. So $\sin 380^\circ = \sin$ ▯.

ii A rotation of 480° has the same rotation as the obtuse angle with measure ▯. So $\sin 480^\circ = \sin$ ▯.

iii A rotation of 580° has the same rotation as the reflex angle with measure ▯. So $\sin 580^\circ = \sin$ ▯$^\circ$.

b Consider the graph of $y = \sin\theta$ for $0 \leq \theta \leq 720^\circ$:
Describe the nature of $y = \sin\theta$.



c The number of degrees it takes for the curve to complete a full cycle is called the period of the function.
Determine the period of $y = \sin\theta$.

2 Consider the curve $y = \sin x$ and state whether the following statements are true or false:

a The graph of $y = \sin x$ is periodic.

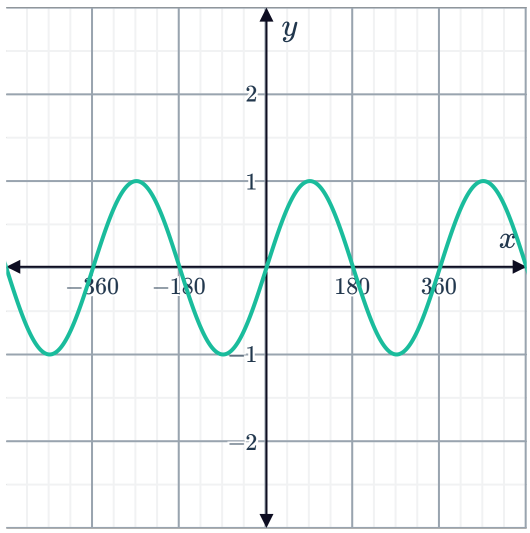
b As x approaches infinity, the height of the graph approaches infinity.

c The graph of $y = \sin x$ is increasing between $x = (-90)^\circ$ and $x = 0^\circ$.

d The graph of $y = \sin x$ is symmetric about the line $x = 0$.

e The graph of $y = \sin x$ is symmetric with respect to the origin.

f The y-values of the graph repeat after a period of 360°.



3 Consider the curve $y = \cos x$ and state wh[redacted] the following statements are true or false:

a The graph of $y = \cos x$ is cyclic.

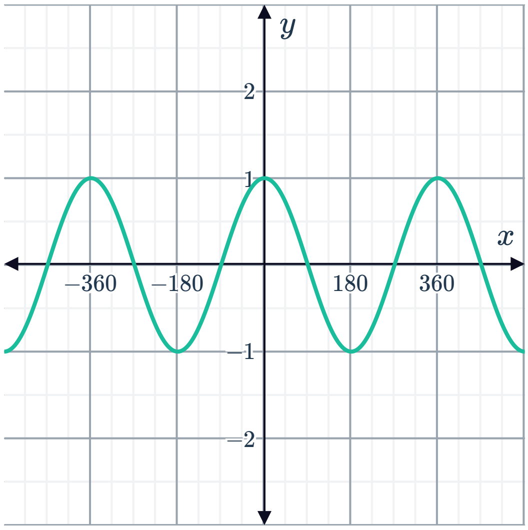
b As x approaches infinity, the height of the graph approaches infinity.

c The graph of $y = \cos x$ is increasing between $x = (-180)^\circ$ and $x = (-90)^\circ$.

d The graph of $y = \cos x$ has reflective symmetry across the line $x = 0$.

e The graph of $y = \cos x$has rotational symmetric with respect to the origin.

f The y-values of the graph repeat after a period of 180°.

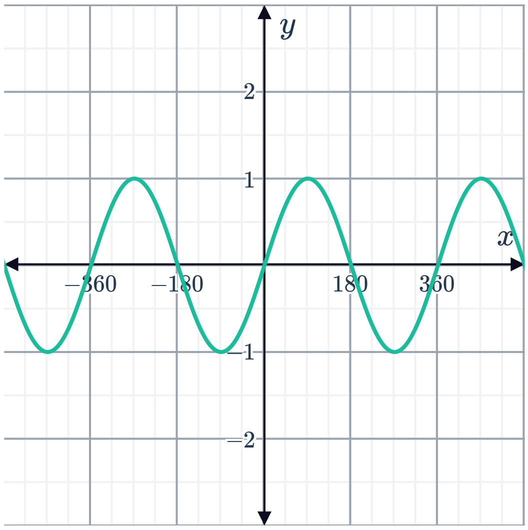


4 Consider the curve $y = \sin x$:

a State the coordinates of the y-intercept.

b Find the maximum y-value.

c Find the minimum y-value.

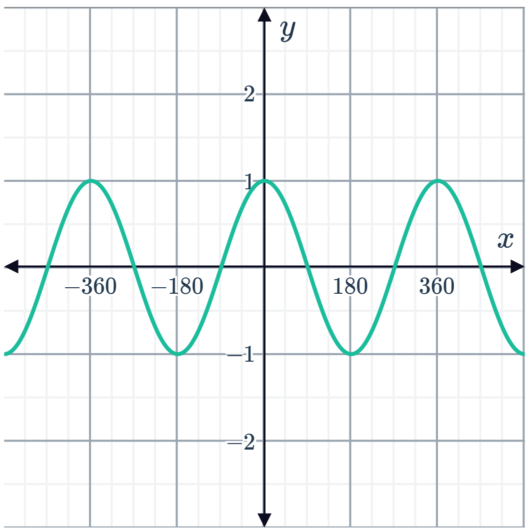


5 Consider the curve $y = \cos x$:

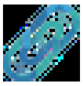
a State the coordinates of the y-intercept.

b Find the maximum y-value.

c Find the minimum y-value.



a Complete the table of values:



x	$0\degree$	$90\degree$	$180\degree$	$270\degree$	$360\degree$
$\cos x$					

b Sketch the graph of $y = \cos x$.

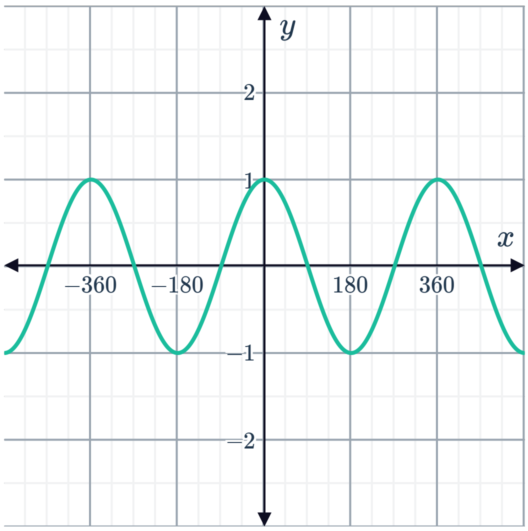
7 Consider the equation $y = \sin x$.

a Complete the table of values:

x	$0\degree$	$90\degree$	$180\degree$	$270\degree$	$360\degree$
$\sin x$					

b Sketch the graph of $y = \sin x$.

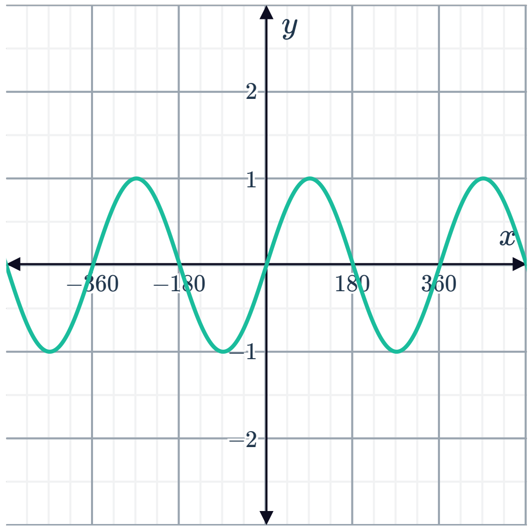
8 Consider the graph of $y = \cos x$:



a Is $\cos 162\degree$ positive or negative?

b In which quadrant does an angle with measure $162\degree$ lie in?

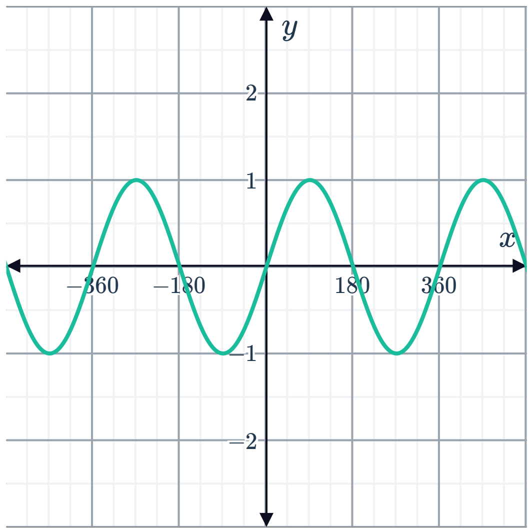
9 Consider the graph of $y = \sin x$:



a Is $\sin 200^\circ$ positive or negative?

b In which quadrant does an angle with measure 200° lie in?

10 Consider the curve $y = \sin x$:



a If one cycle of the graph of $y = \sin x$ starts at $x = 0^\circ$, when does the next cycle start?

b Find the x-value of the x-intercept in the following regions:

i $0^\circ < x < 360^\circ$

ii $(-360)^\circ < x < 0^\circ$

c True or false: As x approaches infinity, the graph of $y = \sin x$ stays between $y = -1$ and $y = 1$.

d State whether the graph of $y = \sin x$ increases or decreases in the following regions:

i $90^\circ < x < 270^\circ$

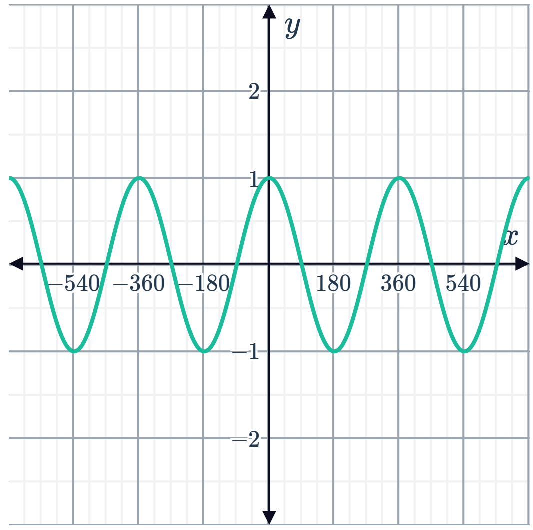
ii $(-270)^\circ < x < (-90)^\circ$

iii $(-450)^\circ < x < (-270)^\circ$

iv $(-90)^\circ < x < 90^\circ$

v $270^\circ < x < 450^\circ$

11 Consider the curve $y = \cos x$:



a If one cycle of the graph of $y = \cos x$ starts at $x = (-90)^\circ$, when does the next cycle start?

b Find the x-values of the x-intercepts in the following regions:

i $0^\circ < x < 360^\circ$

ii $(-360)^\circ < x < 0^\circ$

c True or false: As x approaches infinity, the graph of $y = \cos x$ stays between $y = -1$ and $y = 1$.

d State whether the graph of $y = \cos x$ increases or decreases in the following regions:

i $0^\circ < x < 180^\circ$

ii $(-180)^\circ < x < 0^\circ$

iii $180^\circ < x < 360^\circ$

iv $(-360)^\circ < x < (-180)^\circ$